Manual

Sample title for a manual

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Submission date:	October 19, 2024

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Acronyms

1 Introduction

The NVIDIA Jetson Nano J1010 is a small, powerful AI computing device. This manual provides comprehensive details on the hardware, setup, and usage to help developers and users leverage the power of Jetson Nano to run AI frameworks and applications like image classification, object detection, and speech processing.

Besides the Jetson module, reComputer J1010 also includes J101 carrier board with onboard microSD card slot, 1USB 3.0, 2USB2.0, HDMI, M.2 Key E for WiFI, LTE and Bluetooth, RTC, Raspberry Pi GPIO 40-pin, and so on, as well as a heatsink, and aluminum case. The device has been pre-installed Jetpack 4.6.1, just plug in a USB C 5V/3A power supply, keyboard, mouse, and ethernet cable, you are ready to start your embedded AI journey! If you need more USB 3.0 and onboard M.2 key M for attaching SSD, you can choose reComputer J1020.

2 Hardware Overview

2.1 Specifications

- CPU: Quad-core ARM Cortex-A57
- GPU: 128-core NVIDIA Maxwell
- Memory: 4 GB LPDDR4
- Storage: MicroSD (minimum 16 GB recommended)
- IO: 4 USB 3.0, HDMI, Gigabit Ethernet, 40-pin GPIO header
- Power: 5V/4A DC or Micro-USB

2.2 Physical Dimensions

The Jetson Nano J1010 has the following physical dimensions:

- Dimensions: 100mm x 80mm
- Weight: 100g

3 Sketch of Hardware Ports and Connectors

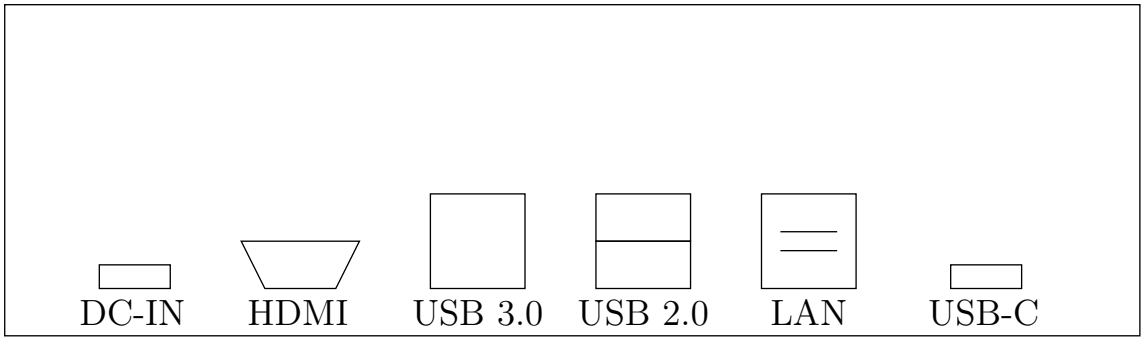

Figure 3.1: Sketch of Jetson Nano Ports and Connectors

4 List Of Parts

The following is a list of components included in the box with the Jetson Nano J1010, as well as accessories required but not included for initial setup:

Included in the Box

- reComputer J1010, including:
    - 4G Jetson Nano module x1
    - J1010 carrier board x1

Required Accessories (Not Included)

- USB Keyboard and mouse
- Display screen
- Type-C power cable and power supply

Ensure that all parts are present before starting the setup process. You will need to source the required accessories separately to power on and configure your Jetson Nano system.

5 Installation and Set Up

5.1 Required Components

To set up your Jetson Nano J1010, you will need:

- NVIDIA Jetson Nano J1010 board
- MicroSD card (16GB or larger) with pre-installed JetPack SDK
- USB keyboard and mouse
- HDMI monitor
- 5V power supply (4A recommended for maximum performance)

5.2 Installing the Operating System

1. Download the JetPack SDK from NVIDIA JetPack SDK.
2. Flash the image to a MicroSD card using tools like Etcher.
3. Insert the MicroSD card into the slot on the Jetson Nano J1010.
4. Connect the HDMI monitor, keyboard, and mouse.
5. Power on the device by connecting the 5V power supply.

5.3 Ethernet Configuration

- Connect a standard Ethernet cable to the Gigabit Ethernet port.
- By default, the system will acquire an IP address via DHCP. To set a static IP, modify the `/etc/network/interfaces` file.

5.4 Wi-Fi Setup (Optional)

1. Plug in a USB Wi-Fi adapter compatible with Jetson Nano.
2. Run the following commands to install the necessary packages:

```
sudo apt-get update
sudo apt-get install network-manager
```

3. Configure Wi-Fi using the Network Manager GUI or by modifying `/etc/network/interfaces`.

6 Running AI Models

6.1 Installing TensorFlow and PyTorch

To run AI models, install TensorFlow or PyTorch:

1. Install TensorFlow:

   ```
   sudo pip3 install tensorflow
   ```

2. Install PyTorch:

   ```
   sudo apt-get install libopenblas-base libopenmpi-dev
   sudo pip3 install torch torchvision
   ```

6.2 Running Pre-Trained Models

You can run pre-trained AI models using NVIDIA's DeepStream or TensorRT libraries:

- Use DeepStream for video analytics and object detection.
- Use TensorRT for optimized, high-performance inference.

7 Power Management

7.1 Power Modes

The Jetson Nano J1010 supports different power modes to balance performance and efficiency:

- Mode 0: Maximum performance (10W)
- Mode 1: Energy-saving mode (5W)

Switch between power modes using the following command:

```
sudo nvpmodel -m <MODE>
```

8 Maintenance and Troubleshooting

8.1 Updating JetPack SDK

- Regularly check for JetPack SDK updates at NVIDIA JetPack SDK page.
- To update, run:

```
sudo apt-get update
sudo apt-get upgrade
```

8.1.1 Device Not Booting

- Ensure the power supply is adequate (5V 4A recommended).
- Reflash the MicroSD card using the latest JetPack version.

8.1.2 Overheating

If the device overheats:

- Ensure proper ventilation.
- Add a fan or heatsink for additional cooling.

9 Compliance and Safety Guidelines

9.1 Compliance with Standards

Ensure that your installation complies with local safety standards and that the hardware meets the necessary certifications.

9.2 Safety Precautions

- Avoid exposing the Jetson Nano J1010 to extreme temperatures or humidity.
- Use a surge protector to protect against voltage fluctuations.